

1. Probability Models

Teaching Team

It's a palindrome!

- Instructor: Sotirios (**Sotos**) Damouras
  - PhD in Statistics (CMU), MSc in Financial Mathematics (Warwick)
    Interested in: Econometrics, Stats Teaching & Consulting, ML/AI, …
    but also: cycling, comic books, cinema, cooking, …
- Our TAs:

TA	TUT	TA	TUT
Tian Wang	01	Yuxiang Jia	06
Kian Dianati	02	Alysa Phoebe Leong	07
Yipeng Zhao	03	Ali Arasteh	08
Ziyi Ye	04	Jiajun Lei	09
Junying Wu	05	Xinyue Zhang	10

- About YOU:
  - What are you studying?

About the Course

- Introduction to the *Theory of Probability*
  - Theoretical basis of Statistics
  - Used *everywhere*: CS, ML, Finance, Life Sciences, …

- Course objective: learn basics of calculus-based probability
  - Translate real-world problems into probability questions
  - Employ rules and techniques for probability calculations
  - Develop probabilistic thinking and reasoning

Course Logistics

- Course administered through Quercus
  - Course materials, Lecture Notes, Quizzes etc
- Textbooks:
  - (main) Introduction to Probability, by Blitzstein & Hwang (B&H)
  - (suppl) Probability and Statistics, by Evans & Rosenthal (E&R)
- Communication
  - For course content questions, bring to tutorials or post on Piazza
  - For personal issues use email or office hours (TU 12-1 & TH 12-2pm @IA4072)
- Missed work policy
  - For any missed work, declare absence on ACORN and email proof to instructor, and weight will be shifted to final

Assessment

Evaluation	***Weight***	***Details***
Weekly Quizzes	15%	• Best 9/11 • In person, during tutorial • Cover previous week's material
Term Test 1	20%	• Cover weeks 1-4 • Date TBD (~ wk 6)
Term Test 2	20%	• Cover weeks 5-8 • Date TBD (~ wk 10)
Final	45%	• Cumulative • Final exam period

Learning Outcomes

By the end of this week, you should be able to:

- Understand concepts & jargon of *probability models*
- Use *naïve set theory* to describe events
- Set up basic (discrete) probability models
- Employ counting for *discrete uniform* probability models
- Apply *rules* to calculate probabilities of *composite* events

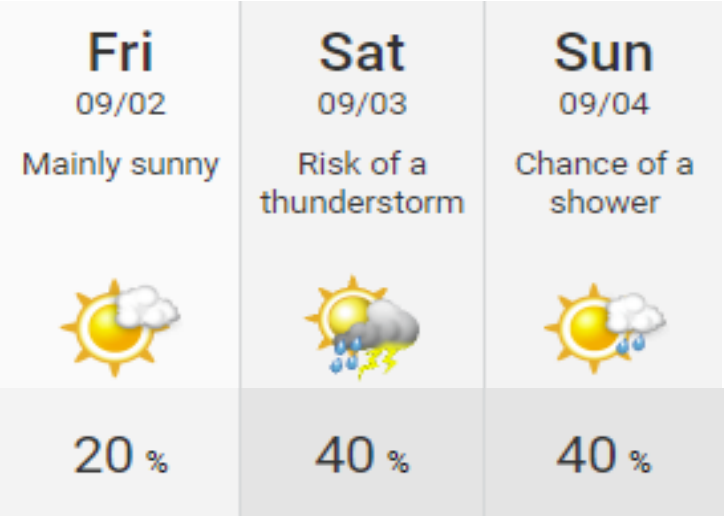
Probabilities

- Many phenomena are uncertain: roll of die, tomorrow's weather
  - Don't know *exactly* what will happen
  - But could know *how likely* is something to happen

- *Probabilities* measure how *likely* is something to happen, in a scale of [0, 1]
  - Value of 1 means it is sure to happen &
    value of 0 means it is not going to happen

Example

- Weather forecast giving daily probability of precipitation



- What can we say about:
  1. Probability of no rain at all in 3 days
  2. Probability of rain in exactly 1 out of 3 days
  3. Probability of rain on Sat, *given* it didn't rain on Fri

- To answer such questions we need a *Theory of Probability*
  - Concepts, terminology, and rules for calculating probabilities

Probability Terminology

- *Random Experiment*: process whose result is "*unknown*"
  - E.g. Roll of a standard (6-sided) die

- *Outcome*: *elementary* result of experiment
  - Outcomes cannot be broken down to simpler results
    - E.g. "Rolling a 1" (vs "Rolling an even number")
  - Typically denoted by lowercase Latin letters, often indexed; E.g., $q, r, s_1, s_2, \ldots, s_n, s_{n+1}, \ldots$

Probability Terminology

- *Sample Space*: collection (*set*) of all possible outcomes
  - E.g. sample space of rolling a die is set {1, 2, 3, 4, 5, 6}, where 1 represents “Rolling a 1”, etc.
  - Typically denoted by S or Ω
- *Event*: arbitrary sub-collection (*subset*) of outcomes
  - E.g. $A = \{2, 4, 6\}$ = “Rolling even number”
  - Typically denoted by “early” uppercase Latin letters
    E.g. $A, B, C, \ldots$
- *Probabilities are assigned to events!*

Set Theory

- *Sets* are (unordered) collections of (distinct) *elements*
  - E.g. set $A = \{a_1, a_2, a_3\}$ has 3 elements
  - Sets can be defined by *listing* or *describing* their elements
    - E.g. $A = \{1, 2, 3\} = \{n \in \mathrm{N} : 1 \leq n \leq 3\} =$ "Roll 3 or less"
- *Universal set* (S) contains *all* possible elements
- *Null* or *empty set* ($\emptyset$) contains *no* elements
- In Probability Theory:
  - Sample space ~ Universal set
  - Outcomes ~ Elements
  - Events ~ Sets

Example

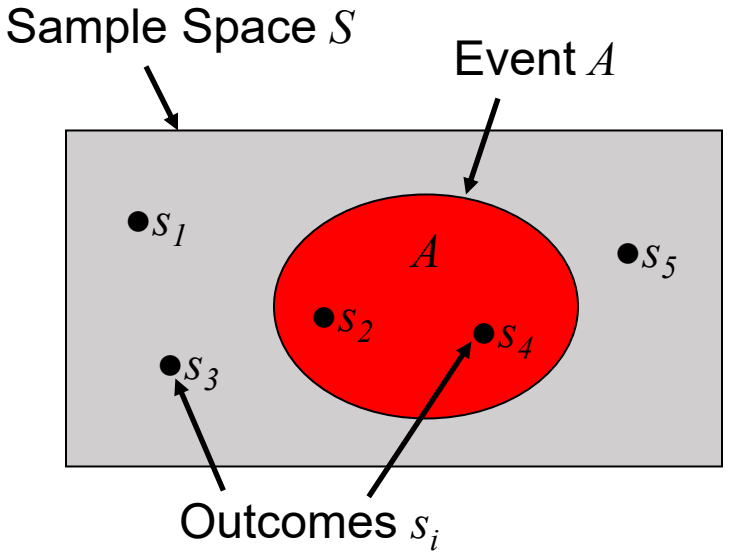
- An insect lays 1-3 eggs, each one being male (M) or female (F)
  - Write a sample space for the “egg-laying” experiment, taking into account the *sex* but *not the order* of the eggs

Example (cont'd)

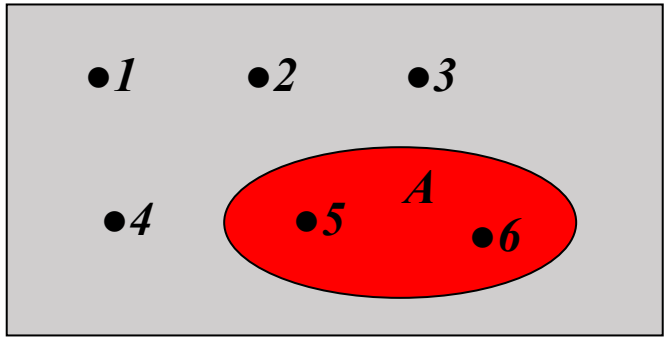
- Rewrite the sample space, considering the *order* in which the sex of the eggs appears.
  - Note: use round brackets (…) to denote *ordered* collections of items (a.k.a. *tuples*)

Venn Diagrams

- Visual representation of sets
  - Sample space is rectangle
  - Outcomes are points in rectangle
  - Events are collections of points, i.e. subsets of rectangle

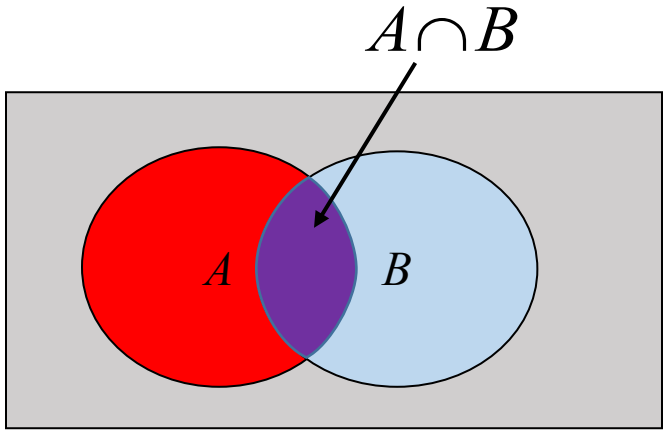


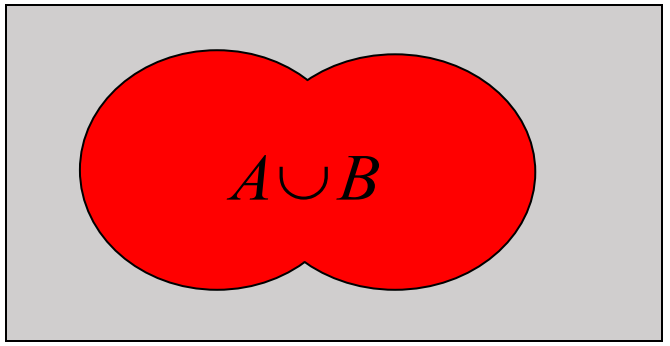
- E.g. Rolling 6-sided die



Set Operations

- *Intersection* of A and B is set of all outcomes that are in *both* events
  - Denoted by $A \cap B$
  - Represents logical AND operation
- *Union* of A and B is set of all outcomes that are in *either* of the two events
  - Denoted by $A \cup B$
  - Represents logical OR operation





Union/Intersection Properties

- Commutative: $A \cap B = B \cap A, \quad A \cup B = B \cup A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C,$

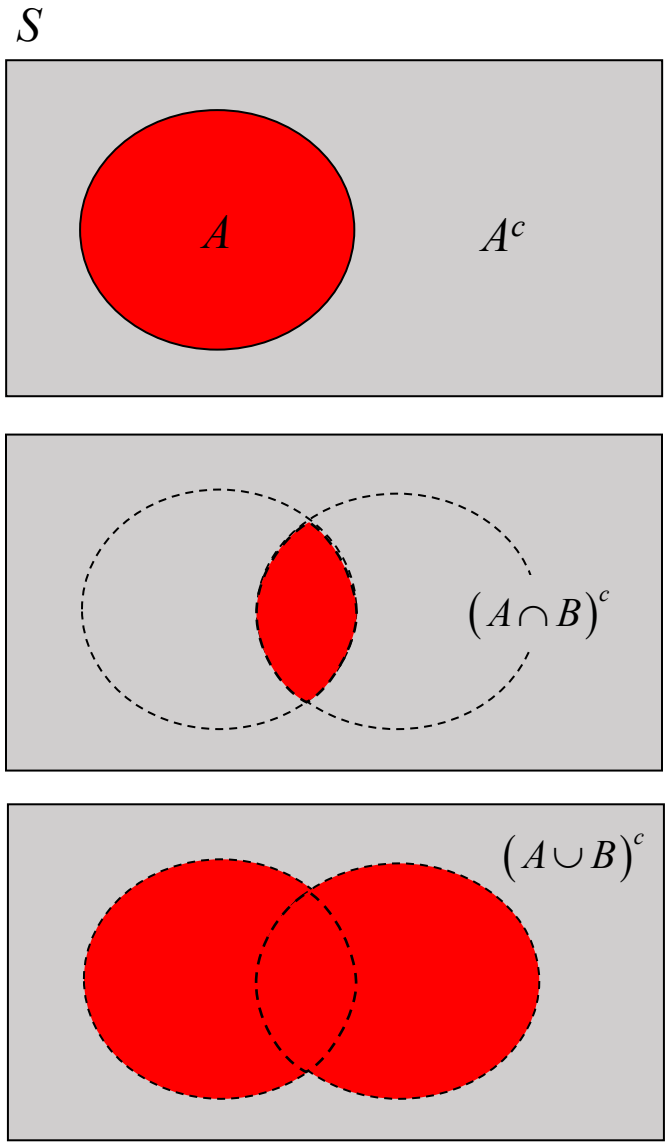
  $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C),$

  $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- Properties apply for larger collections of sets
  - Note: $\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup \cdots \cup A_n$ and similarly for ∩

Complement

- The *Complement* of a set A, denoted by A^c, is the set of all outcomes that are *not* in A
- De Morgan's laws:
  - $(A \cap B)^c = A^c \cup B^c$
  - $(A \cup B)^c = A^c \cap B^c$
- Generally: $\left(\bigcap_{i=1}^{n} A_i\right)^c = A_1^c \cup A_2^c \cup \cdots \cup A_n^c$

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = A_1^c \cap A_2^c \cap \cdots \cap A_n^c$$



Also:

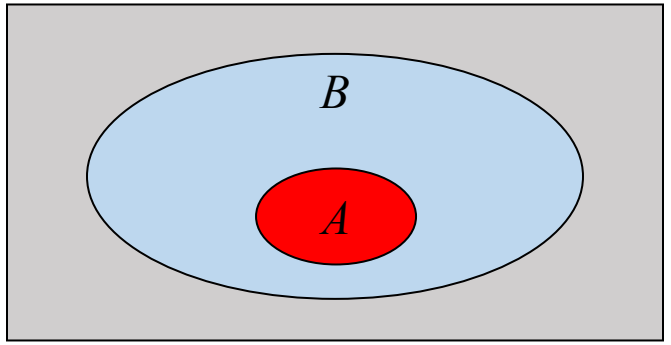
$A \cap A^c = \varnothing$

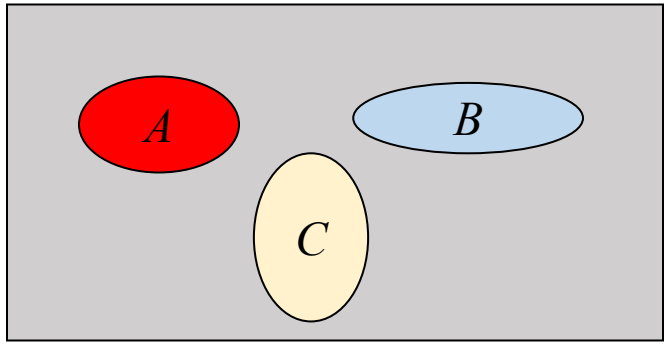
$A \cup A^c = S$

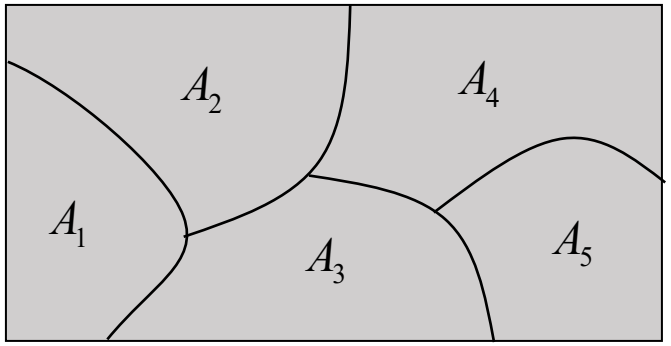
$S^c = \varnothing$

Event Relations

- A is a *subset* of B if every element in A is also in B
  - Denoted by $A \subseteq B$
  - $\emptyset$ is a subset of any set
  - If $A \subseteq B$ and $B \subseteq A \Rightarrow A = B$
- Two or more events are *disjoint* or *mutually exclusive*, if they have no elements in common
- Two or more events form a *partition* if they are disjoint *and* their union is the sample space







Probability Axioms

- *Probability function/measure* is (set) function that takes events and assigns probability values to them
  - Denoted by $P(\cdot)$, where $P(A)$ is probability value of specific event A
- *Probability Axioms*: assigned probabilities (P) must satisfy:
  1. $P(A) \geq 0, \quad \forall A \subseteq S$
  2. $P(S) = 1$
  3. ∀ disjoint $A_1, A_2, A_3, \ldots \Rightarrow P\left(\cup_i A_i\right) = \sum_i P(A_i)$
- *Probability model* consists of triple $(S, \{A, B, \ldots\}, P)$

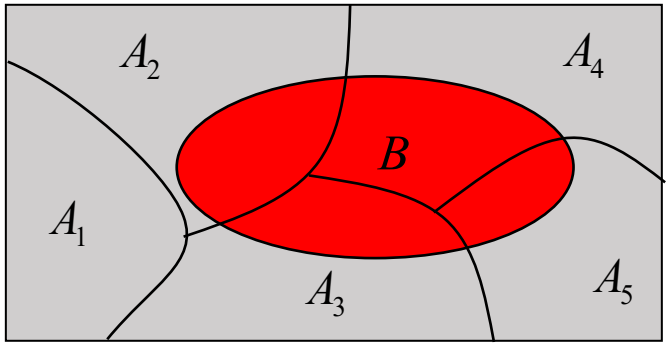
sample space — set of events — probability function

Probability Rules

- *Law of Total Probability*: ∀ partition $A_1, A_2, A_3, \ldots$ & arbitrary event B

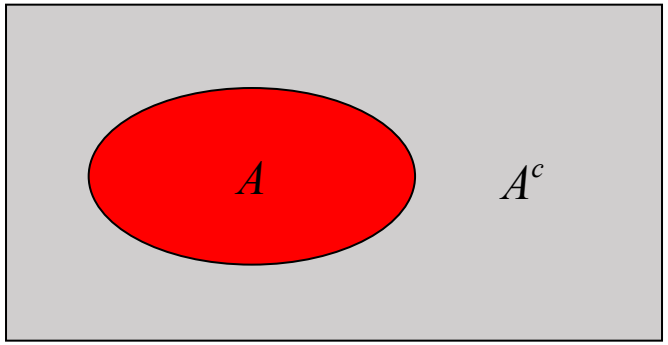
$$P(B) = P(B \cap S) = \sum_i P(B \cap A_i)$$

  - Proof:



- *Complement Rule*: ∀ event A

$$P(A^c) = 1 - P(A)$$



Probability Rules

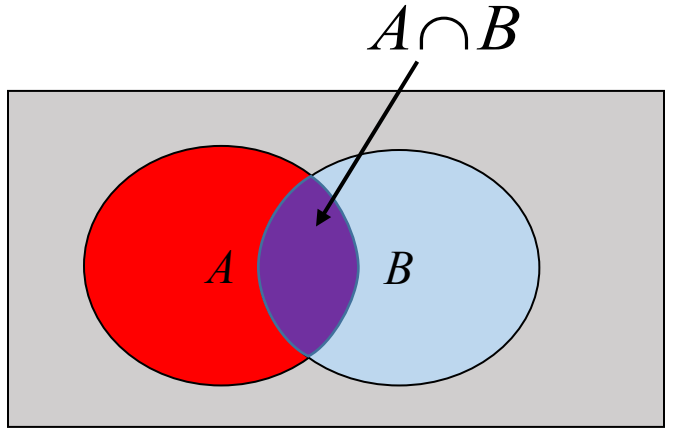
- *Inclusion-exclusion Principle*:

For *any two* events A, B

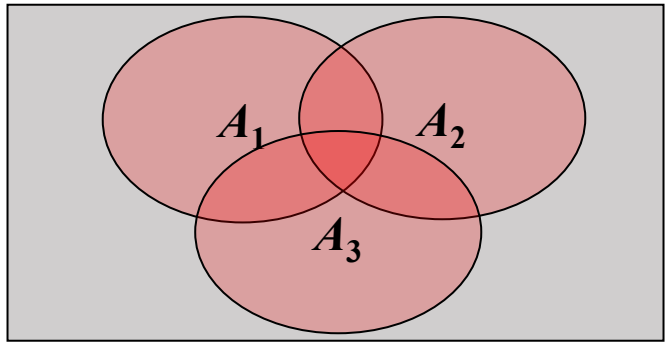
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For *any #n* events $A_1, \ldots, A_n$

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_i P(A_i) -$$
$$- \sum_{i<j} P(A_i \cap A_j)$$
$$+ \sum_{i<j<k} P(A_i \cap A_j \cap A_k) -$$
$$\vdots$$
$$(-1)^{n+1} P(A_1 \cap \cdots \cap A_n)$$



e.g. for n=3

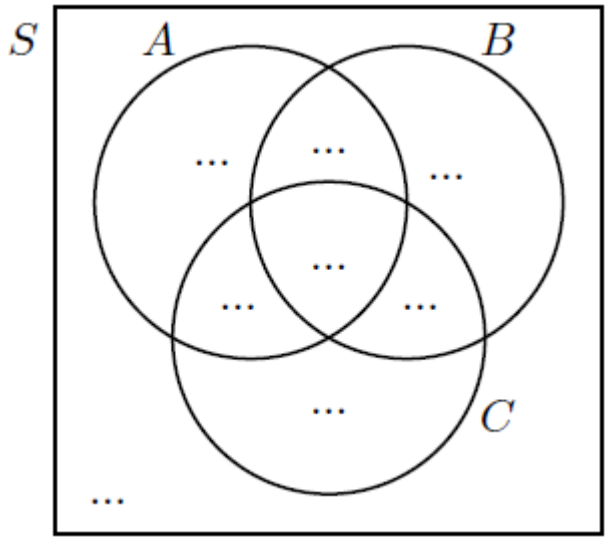


Example

- A student can belong to clubs denoted by A, B, C. Assume:

$$P(A) = P(B) = P(C) = .3$$
$$P(A \cap B) = P(A \cap C) = P(B \cap C) = .2$$
$$P(A \cap B \cap C) = .1$$

  - Fill in the probabilities on the Venn diagram
  - Find the probability that a student belongs to *exactly two* clubs.



Discrete Sample Spaces

- Sample spaces w/ separable outcomes called *discrete*; can be
  - *Finite*: finite # of elements
    - Ex. Roll of a die, toss of a coin
  - *Countably infinite*: elements can be put in 1-to-1 correspondence with natural numbers ($\mathbb{N} \equiv \{1, 2, 3, \ldots\}$)
    - Ex. Number of coin tosses before first head appears
- For discrete S, probability function P is *uniquely determined* by outcome probabilities $P(s_i), \ \forall s_i \in S$

$$A = \{s_1, s_2, \ldots\} \ \Rightarrow \ P(A) = P(s_1) + P(s_2) + \cdots$$

Discrete Uniform Probability

- Finite spaces whose outcomes have *equal probability* are called *discrete uniform* probability spaces
  - $S = \{s_1, ..., s_n\}$, where $n = |S|$ is *size* of S
  - $P(s_i) = 1/|S| > 0, \forall i = 1, \ldots, n$

- For finite uniform probability spaces and any event A,

$$P(A) = \frac{|A|}{|S|} = \frac{\text{\# outomes in } A}{\text{\# outomes in } S}$$

Counting

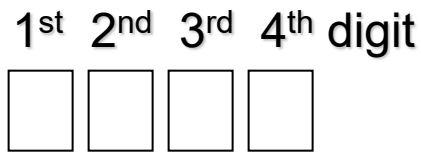
- Counting outcomes is essential for discrete uniform probability
- There are three basic counting rules:
  - *Multiplication*
  - *Permutation*
  - *Combination*
- *Multiplication rule*: with m elements $\{a_1,...,a_m\}$ and n elements $\{b_1,...,b_n\}$, there are $m \times n$ possible *ordered* pairs (a_i,b_j)

Example

- What is the number of sample points for n coin flips?
- What is the number of sample points for n die rolls?
- If S has n elements, how many different subsets can we have?

Permutations

- (Hacking 101) How many 4-digit PIN's are there?



  - How many 4-digit PIN's with *distinct (non repeated) digits* are there?

- *Permutation*: *ordered* arrangement or k objects, chosen *without replacement* from n possible objects.
  - # of possible permutations is: $P_k^n = n \times (n-1) \times (n-2) \times \cdots \times (n-k+1)$

Factorials

- *Factorial* function: $n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$
  - E.g. $4! = 4 \times 3 \times 2 \times 1 = 24$
  - Defined for any $n \geq 0$
  - By convention $0!=1$
  - Stirling's approximation $n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n$
- Permutation formula using factorials:

$$P_k^n = \frac{n!}{(n-k)!}$$

Combinations

- In Lotto 6$_{49}$ you pick 6 out of 49 numbers. How many different tickets are there?



- *Combination*: *unordered* collection of k objects chosen *without repetition* from n possible objects
  - # possible combinations is $C_k^n = \binom{n}{k} = \frac{P_k^n}{k!} = \frac{n!}{k!(n-k)!}$